

Repolyx

MVP Documentation — AI-Powered Developer Assistant for
GitHub Repository Understanding

Version Status Type

1.0 — MVPPlanningTechnical Specification

Project Overview

Repolyx is an AI-powered developer assistant that helps developers understand and manage GitHub repositories more easily.

Instead of manually searching through hundreds of files, developers can connect their GitHub repository and ask questions in simple language. The AI helps explain the project structure, reviews pull requests, helps debug errors, and gives useful suggestions.

The main goal of Repolyx is to make software development faster, easier, and more organized using AI.

Modern developer tools are moving toward repository-aware AI systems and intelligent workflows, which is the same direction Repolyx follows.

Problem Statement

When developers work on large projects, they often face problems like:

- Understanding old code
- Finding where features are implemented
- Debugging errors

- Reviewing pull requests
- Onboarding into new projects

These tasks take a lot of time.

Repolyx solves this problem by creating an AI assistant that understands the repository and helps developers quickly find answers.

Main Goal of MVP

Build a smart AI assistant that understands GitHub repositories and helps developers work faster.

This project is not trying to replace developers or build a fully autonomous AI engineer.

The focus is:

- Repository understanding
- AI-powered assistance
- Developer productivity

Target Users

- Full stack developers
 - Startup teams
 - Freelancers
 - Junior developers
 - Students learning real-world projects
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Core Features

01

GitHub Login

Sign in using GitHub. Repositories become visible and can be connected securely.

02

Repository Connection

Select a repository from GitHub. The system fetches files, scans folders, and reads project structure.

03

Repository Intelligence

AI analyzes folder structure, APIs, dependencies, and auth flow — then generates a project summary.

04

AI Repository Chat

Ask questions about the repository in natural language. Uses RAG with vector search and embeddings.

05

AI Pull Request Review

Paste PRs or changed files. AI detects bugs, missing validation, bad practices, and security risks.

06

AI Debug Assistant

Paste terminal errors, logs, or stack traces. AI explains the error, possible cause, and suggested fix.

07

Dashboard

Shows connected repositories, total reviews, AI activity, recent PRs, and repository insights.

08

Documentation Generator

Automatically generate README files, API documentation, setup guides, and endpoint explanations.

1. GitHub Login

Users can sign in using GitHub. After login, repositories become visible and users can connect repositories securely.

GitHub integration makes the application feel practical and real-world.

2. Repository Connection

Users can select a repository from their GitHub account. The system then fetches repository files, scans folders, and reads project structure.

```
/frontend
/backend
/routes
/controllers
```

This helps the AI understand the project.

3. Repository Intelligence

This is the main feature of Repolyx. The AI analyzes:

- Folder structure
- APIs
- package.json
- Dependencies
- Authentication flow

Then it generates a simple project summary.

"This project uses Next.js frontend with Express backend and JWT authentication."

This makes large projects easier to understand.

4. AI Repository Chat

Users can ask questions about the repository.

Example questions:

- "Where is authentication handled?"
- "Which API handles payments?"
- "Explain backend flow"

The AI searches repository context and gives answers in simple language.

This feature works using vector search, embeddings, and repository context retrieval. This technique is called RAG (Retrieval-Augmented Generation).

5. AI Pull Request Review

Users can paste pull requests, commit changes, or changed files. The AI reviews the code and detects:

- Possible bugs
- Missing validation
- Bad coding practices
- Security risks

"This API route does not validate user input."

This feature is inspired by modern AI review systems like CodeRabbit.

6. AI Debug Assistant

Users can paste terminal errors, logs, or stack traces. The AI explains what the error means, possible cause, and suggested fix.

EXAMPLE ERROR

MongoNetworkError

"Database connection string may be invalid."

This helps beginners understand errors faster.

7. Dashboard

The dashboard shows connected repositories, total reviews, AI activity, recent pull requests, and repository insights.

The UI should feel clean, modern, and developer-focused. Inspired by tools like GitHub, Linear, Cursor, and Vercel.

8. Documentation Generator

The AI can automatically generate:

- README files
- API documentation

- Setup guides
- Endpoint explanations

This saves developer time.

Simple Workflow

- 1 User logs in with GitHub.
 - 2 User selects repository.
 - 3 System scans repository files.
 - 4 Repository data gets stored.
 - 5 AI understands repository structure.
 - 6 User can ask questions, review PRs, and debug errors.
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Tech Stack

FRONTEND

Next.js

Tailwind CSS

ShadCN UI

BACKEND

Node.js

Express.js

AI INTEGRATION

OpenRouter

Claude

Gemini

GPT Models

DATABASE

PostgreSQL

VECTOR DATABASE

Qdrant

Used for embeddings, semantic search,
and repository context.

GITHUB INTEGRATION

GitHub OAuth

Octokit API

Simple System Architecture

Frontend



Backend API



AI Service



Vector Database



AI Models

Why This Project Is Strong

This project demonstrates:

- Full stack development
- GitHub API integration
- AI integration
- Repository intelligence
- Backend engineering
- Practical problem solving

This is much stronger than basic portfolio projects like todo apps, weather apps, or simple CRUD projects.

Future Scope

Future improvements can include:

AI Code Suggestions

Automatic Bug Fixing

Team Collaboration

Slack Integration

CI/CD Integration

Advanced Analytics

Conclusion

Repolyx is an AI-powered engineering assistant that helps developers understand repositories, review code, and debug problems faster.

The project focuses on practical AI integration and real developer workflows instead of complicated AI research systems.

"Make software development easier with intelligent repository understanding."

